

Illustrate ML concept & suitable examples.

⇒ ML is a subset of AI that involves development of algorithms & models that enable comp. to learn patterns & make predictions without explicit programming.

Fundamental concepts of ML:-

1) Supervised Learning:-

A type of ML where alg is trained on a labeled dataset, consisting of Inp/output pairs.

Eg:- Predicting house price based on features like size, location, & no. of bedrooms using a dataset

2) Unsupervised Learning:-

algorithm is given unlabeled data & must find patterns or relationship within it.

Eg:- clustering customer data to identify groups with similar purchasing behaviour

3) Feature Extraction:-

The process of selecting or transforming raw data into format suitable for model training.

2. Eg:- Counting the frequency of words from text data.

3) Cross-validation:-

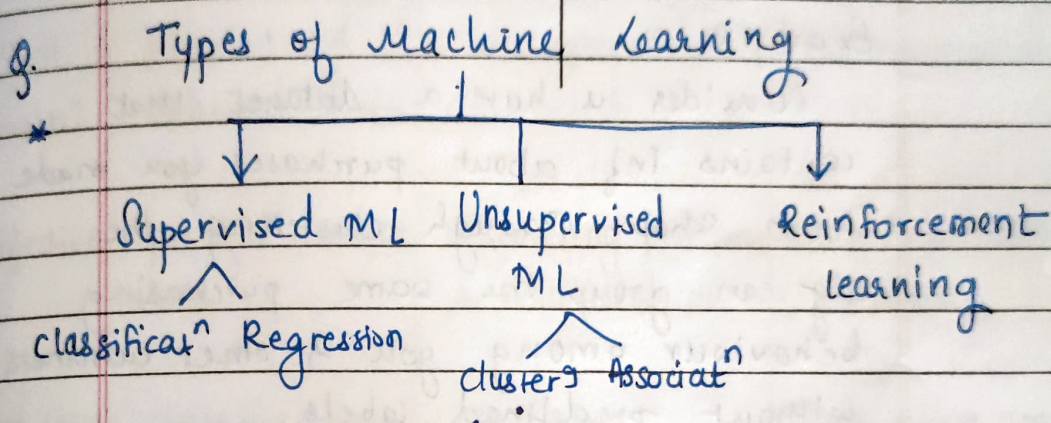
Technique for assessing a model's performance by dividing dataset into multiple subsets for training & testing.

Eg:- K-fold cross-validation, splitting the data into k subsets, training the model on $k-1$ subsets, & testing remaining subsets.

Q. classic Machine	Adaptive Machine
- It relies on predefined algorithms & fixed model	- adapts & evolves based on new data.
- Trained on static datasets, requiring periodic updates.	- Continuously learns & updates its knowledge as new data comes.
- less flexible	- well-suited & flexible
- might struggle with complex, non-linear relationships	- potentially handle complex, non-linear or .

- Requires manual intervention to update model with new knowledge

- Automatically adjust to new data, providing real-time learning capabilities.



1. Supervised

- The machine learns by labeled data.
- Labelled datasets have both input & output parameters.

Scenario:-

Consider a scenario, where u have to build image classifier to differentiate between cat or dog. If you feed the datasets of dogs & cats labelled imgs to alg, the machine will learn to classify betⁿ dogs & cats. When we input dog or cat images that it has never seen before, it will used learned alg. & predict. This is image classification.

2} Unsupervised:-

- technique in which an algorithm discovers patterns & relationships using unlabeled data
- goal is to discover hidden patterns, similarities, which can be used for data visualization, exploration, more.

Example:-

consider u have a dataset that contains inf about purchases you made from shop. Through clustering, the alg can group the same purchasing behaviour among you & other customers without predefined labels.

- This type of inf can help businesses get target customers.

3} Reinforcement:-

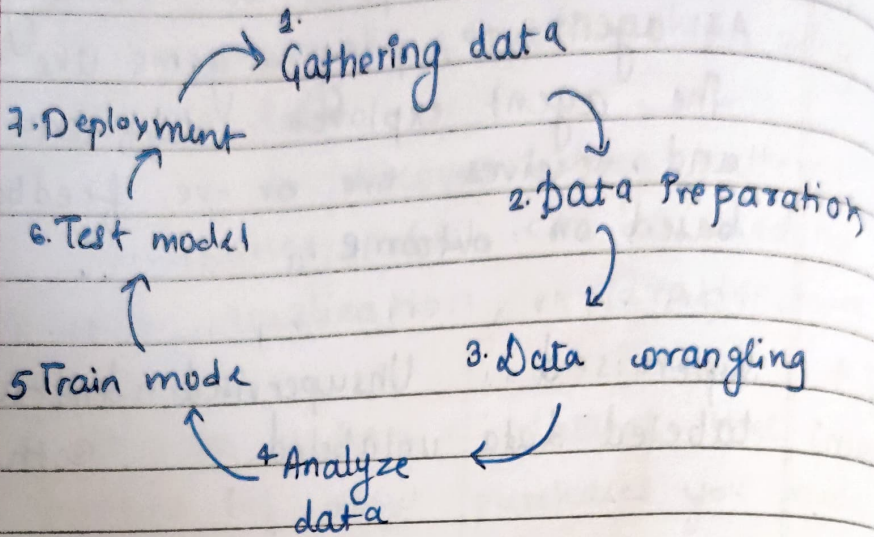
- a learning method that interacts with environment by producing actions & discovering errors
- Each time we feed data, it learns & add the data to their knowledge which is training data.
- So, model more it learns, the better it gets trained.
- model learns from interacting with an environment & receiving feedback in form of rewards & penalties.

Eg:-

Consider that you are training an AI agent to play a game like chess. The agent explores different moves and receives +ve or -ve feedback based on outcome.

Q. 4. Training	Supervised labeled data	Unsupervised unlabeled	Semi-supervised. Both
Objective	predict ^g output labels	discovering patterns & relationships	Balancing labeled & un data.
use cases	classification, Regression	clustering, Association	Tasks where acquiring labeled data is challenging
Adv:-	- high accuracy - clear objective, task-specific	versatile for discover ^g patterns - useful for dimensionality reduction	- cost effective! - flexibility in utilizing available lab/unlab data
Disadv:-	- struggle with unstructured d. - limited adaptability to unknown patterns	- Subjective evaluat ⁿ . - challenges in assess ^g model's performance	- success depends on effective use of data - complex

Q 5. Machine Learning Life Cycle.



→ 1. Gathering data:-

The obj is to identify & obtained all data related objects. The more the data, the more accurate predictions.

i) identify data sources

ii) collect data

iii) collect data from diff sources.

2. Data preparatⁿ:-

In this state, we put our data into a suitable place. Data is prepared by using data exploration

& Data preprocessing.

3. Data wrangling:-

- cleaning & removing errors.

Includes - missing value, duplicates, invalid data, errors.

4) Analyze Data:-

In this step, clean & prepared data is passed for analyzation.

Analyze data by using diff ML techniques like classificatⁿ, reg, clustering.

5) Train model:-

- use dataset to train model so that we can understand various patterns rules, features using ML alg.
- Training model improve performance for better outcomes

6) Test model

We check accuracy provided by training dataset.

7) Deployment.

We deploy our build model in the real world system,

AL	ML	DL
1) It is a broader concept. 2) encompasses machines carry out tasks in intelligent manner. 3) Requires labeled & unlabeled data. 4) Application:- NLP, Robotics. 5) Rule-based learning, & learn from experience.	1) It is a subset of AI. 2) focuses on develop of alg. that enables comp. to learn patterns. 3) depends on labeled data for training & unlabeled for generalization. 4) Image recognition, Speech recognition. 5) emphasizes learn from data without explicit programming.	1) It is specialized form of ML. 2) involves neural networks with multiple layer to model complex patterns. 3) requires labeled data for training deep neural network. 4) NLP, Speech & image recognition. 5) relies heavily on neural network architectures.

Q 7. *

• Training

Testing.

Objective:- Teach the alg to recognize patterns or relationships within a dataset.

Process:- feed the alg with a labeled dataset (inp-out pairs)

Outcome:- alg adjusts its parameters its parameters to learn from data for accurate predictions.

• Testing:-

Objective:-

Evaluate model's performance on new, unseen data.

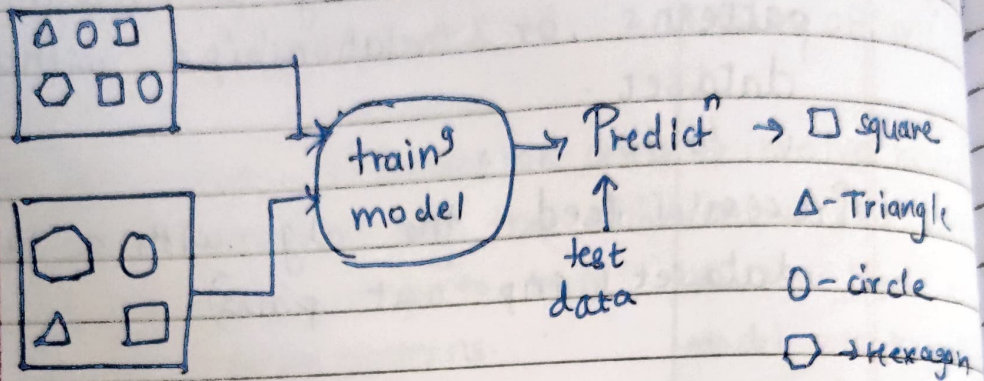
Process:-

Use a separate dataset (Not used during training) to assess model's predictions.

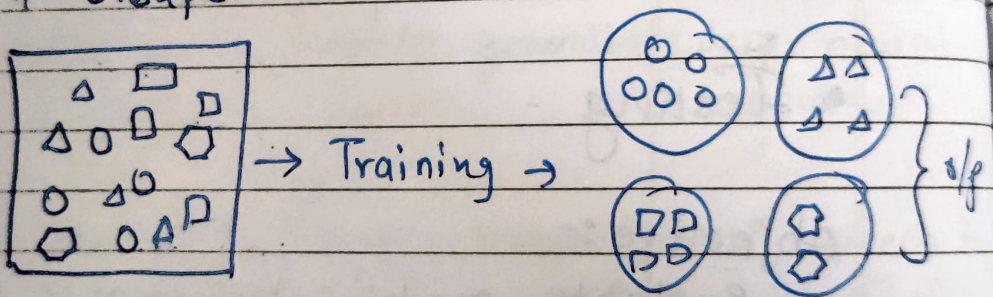
Outcome:-

measure how well the model generalizes to new data, indicating its effectiveness

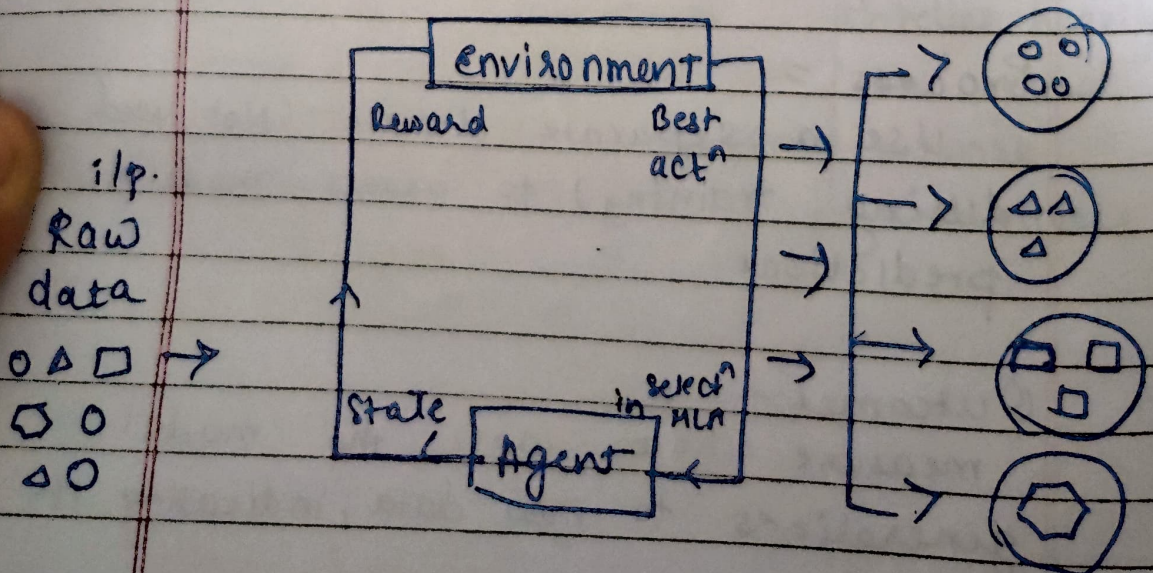
1) Supervised



2) Unsupervised



3) Reinforcement.



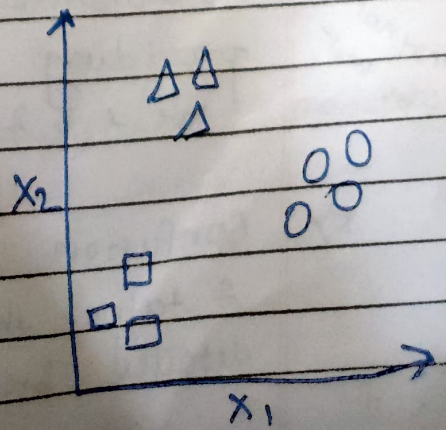
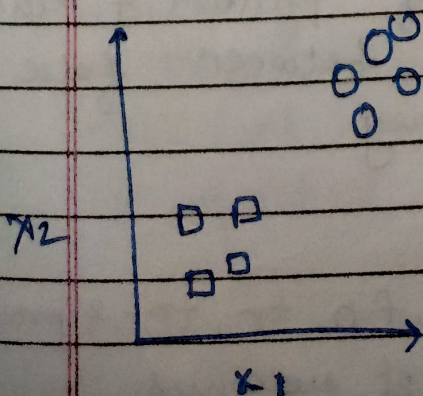


coms eg.

Binary classificatⁿ

Multi class

- | | |
|---|---|
| - has two classes
(+ve & -ve) | - has more than 2
classes |
| - It outputs a single
probability. | - It provides prob.
distribut ⁿ for each class. |
| - Evaluat ⁿ metrics -
accuracy, precision,
recall, F1 score. | - Evaluat ⁿ - s
"
" |
| - Generally require
simpler models | - may require more
complex models. |
| - Eg. spam detect ⁿ ,
fraud detect ⁿ | - handwriting recognition,
object recognit ⁿ . |
| - Alg used r logistic
reg, k-nearest neighs
Naive Bayes. | - Alg - Decision trees,
Random forest,
Naive Bayes |



Q. Performance metrics & related terms for binary classificatⁿ.

1) Accuracy:-

mislead^g in
imbalanced
dataset.

How often the model gets things right overall. It measures proportion of correctly classified instances out of all instances.

2) Precision:-

may not
be informative

Indicates accuracy of positive predictions calc. as true positives divided by false positive & true positives.

$$\frac{TP}{TP + FP}$$

3) Recall:-

may
not be
best

represent the ability of model to capture all relevant instances (calc. as true positives div by true positive and false neg.

4) F1 score:-

may not
be best

Harmonic mean of precision & recall, providing balance between false pos^{ve}'s & false neg.

5) Confusion matrix:-

A table show^s f.p, f.n, tn, tp & provide^d detailed breakdown of outcomes.

6) ROC (Receiver Operat^g char):-
Adjust^g to catch more trues without
increas^g false trues.

7) AUC - ROC (Area under the ROC curve):-
A measure of how good the model is
at distinguish^g betⁿ true & -ve cases

8) Specificity:-
measures accuracy of -ve predictions
calc as $\frac{T.N}{T.N + F.P}$

9) Diff. One -vs rest & One vs one strategies in
multiclass classificatⁿ.

One -vs rest	One vs One
- Each class vs rest	- Pairwise classificat ⁿ .
- NO. of classifiers - $N(\text{no. of classes})$	- $N*(N-1)/2$
- low computational intensity.	- high computational intensity
- Simple	- Complex
- efficient for large no. of classes	- suitable for moderate class counts.

Scenario :- Large No. of classes.

OVR :- often preferable when dealing with large no. of classes because training N classifiers is computationally efficient.

OVO :- might become impractical due to increase in classifiers.

Scenario :- Class Imbalance

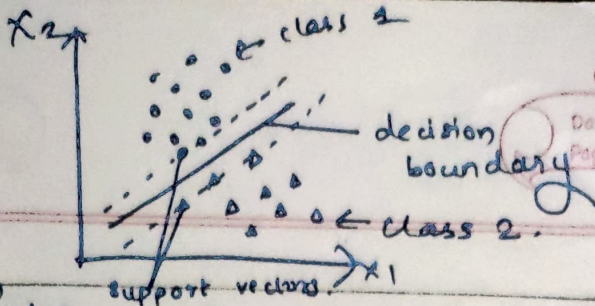
OVR :- can be impacted by imbalances

OVO :- Robust in scenarios with significant class imbalance, as each classifier focuses on distinguishing between specific class pairs.

SVM $\rightarrow$

finds hyperplane that best separates data points into different classes.

It can handle linear and non linear classification.



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SVM - linear regression classification problems.

⇒ Support Vector Machines (SVM)s

1) Linear Reg:-

The goal is to find a line that best fits data.

SVM for regression, often called - SVR (R-regressⁿ), aims to minimize errors, still maintaining margin of tolerance.

- kernel func. can be used to handle non-linear relationships in SVR, extends it to non-linear regression problems.

2) Classification:-

- widely known for its appli in classifi tasks.

- for binary classificatⁿ, SVM aims to find a hyperplane that separates data into two classes, maximizing margin betⁿ them.

- Data points lie closest to decision boundary.

- Play a key role in determin^g position & orientatⁿ of boundary.

SVM in Healthcare Industries.

⇒ Disease Diagnosis:-

- SVMs can assist in diagnosing diseases based on patient data.
- for eg, can analyze medical records, lab results, patient history to classify individuals as either healthy or not.

2) Cancer Classification:-

- effective in cancer classⁿ task.
- By analyzing genetic & molecular data SVMs can help identify patterns associated with diff. types of cancer, aid^g in early detectⁿ & treatment plans.

3) Image analysis.

- tumor detectⁿ in radiological img.
- excel in distinguish^g betⁿ diff. structures within images.

4) Fraud detectⁿ & Billing:-

- can assist in detect^g fraudulent activities related to healthcare billing
- identify irregularities.

Drug discovery:-

- play a role in predicting efficacy of potential drug compounds.
- By analyzing molecular & chemical data, it helps identify promising drug candidates & assess impact on specific diseases.

Kernel Methods:-

Laplace kernel - used to measure similarity betⁿ 2 input vectors

$$k(x, y) = \exp(-\gamma \|x - y\|)$$

SVMs use kernel methods to transform input data into a higher-dimensional feature space, which makes it simpler to distinguish betⁿ classes or generate predictions.

- Kernel funcⁿ in SVMs is essential in determining decision boundary that divides various classes.

- In order to calculate the degree of similarity betⁿ any two points in feature space, kernel funcⁿ computes their dot product.

- Kernel methods r powerful technique in solv^g classificatⁿ & regression problem

- widely used in ML as they can handle complex data structures

funcⁿ - Linear - dot product betⁿ input vectors $k(x, y) = x \cdot y$

Polynomial - nonlinear kernel funcⁿ employs polynomial funcⁿ

Gaussian (RBF) - uses gaussian funⁿ $k(x, y) = \exp(-\gamma \|x - y\|^2)$

decision boundary

degree of similarity

classification reg.

data structure

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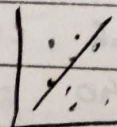
1) Regression & its Types.

Reg. is predictive modeling technique used when output variable is continuous or numerical.

It is a statistical model method used to model the relationship betⁿ a dependent var & one or more independent variable. widely used in various fields such as economics, finance, social sciences, & natural sciences.

✓ Types :-

1) Linear Regression :-



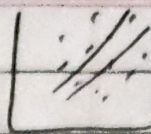
- simplest form that assumes a linear relationship betⁿ independent & dependent variable. It fits a straight line to the data points that presents rel. betⁿ variables

eg

$$y = B_0 + B_1x + E$$

$\downarrow$ $\uparrow$ $\uparrow$

dependent var independent var error term



2) Multiple linear Regression:-
extends l.r to include multiple independent var.

models rel. betⁿ dependent var & one or more ind var.

$$y = B_0 + B_1x_1 + B_2x_2 + \dots + B_nx_n + e.$$

3) Polynomial Reg

→ used when relationship betⁿ ind & dep var is curvilinear rather than linear.

It fits polynomial func. to data points.

$$y = B_0 + B_1x + B_2x^2 + e$$

4) Logistic Reg

used when dep var is categorical.

It models probability of occurrence of an event by fitting a logistic curve to data.

$$\log\left(\frac{p}{1-p}\right) = B_0 + B_1x_1 + B_2x_2 + \dots + B_nx_n$$

5) SVM Regression

The goal is to find straight line that best fits the data. used in regression for minimizing errors & stills have margin of tolerance. It uses kernel funcⁿ to have non-linearity issues sol^d.

Q3. Least Sq. Method

→ popular technique used in regression analysis to estimate parameters of a linear model by minimizing sum of squared diff betⁿ observed and predicted values of dependent var.

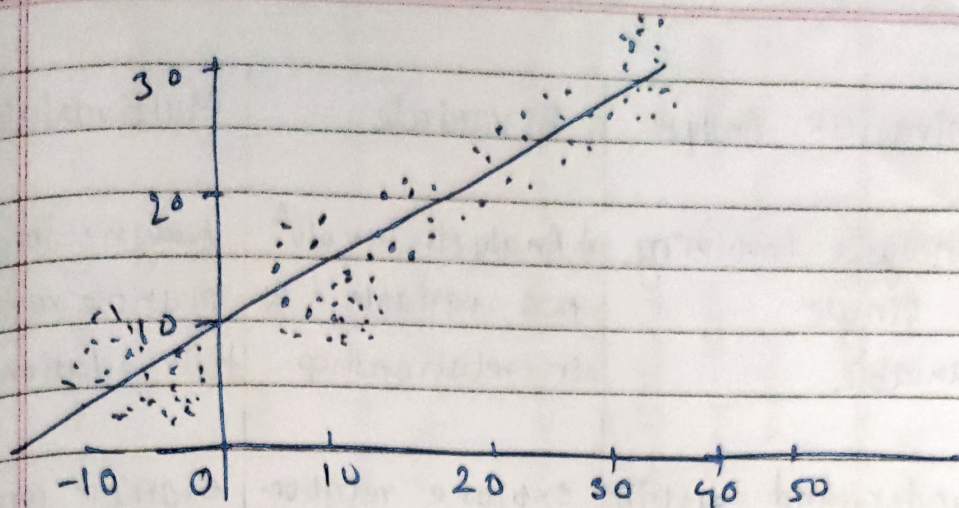
There are two basic categories of least-sq. problems:-

- Ordinary or linear least squares
- nonlinear least squares.

These depend upon linearity or nonlinearity of residuals. The linear problems are often seen in regression analysis in statistics.

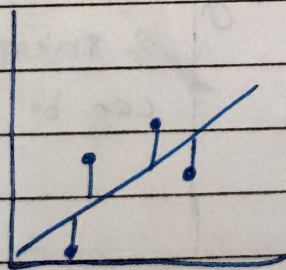
Non-linear problems are generally used in iterative method of refinement in which model is approximated to linear one with each iteration.

In linear reg, the line of best fit is a straight line as shown in diag:-

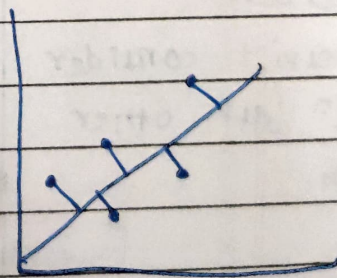


The given pts are to be minimized by method of reducing residuals or offsets of each pt from line.

The vertical offset is used in surface, polynomial, & hyperplane problems, while perpendicular offsets are utilized in common practice.



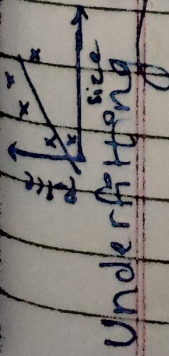
Vertical
offsets



Perpendicular
offsets

The eqⁿ of least sq line is $y = a + bx$.

Univariate Analysis	Bivariate	Multivariate
1) Analysis involving a single variable	1) Analysis involving two variables & their relationship	Analysis involving multiple variables & their relationship
2) understand distribution of single variable	explore relationships between two variables	examine complex relationships among multiple variables
3) eg Histograms Box plots	eg Scatter plots - chi-square test	eg Multiple reg. - cluster analysis - PCA.
4) Adv - simple & easy	- helps identify patterns	- captures complex interactions
5) Disadv doesn't consider relationship with other variables.	- may oversimplify complex interactions	- Data preparation & interpretation can be challenging



Underfitting

Model is too simple & fails to capture underlying patterns in data

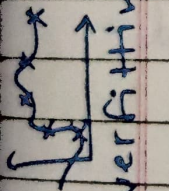
char:- high training error

- fails to capture underlying patterns

- over simplified model

- poor performance on both training & test data

- High bias, low variance.



Overfitting

Model is too complex and captures noise in the data.

- low training error

- captures noise and fluctuation in data

- Overly complex model

- excellent performance on training, poor on testing.

- low bias, high variance

Variance

degree of fluctuation or variability in model predictions.

- model predictions vary significantly for diff datasets.

- sensitive to variations in training data

- may lead to overfitting or underfitting if not balanced

- can cause instability & lack of generalization

Bias

error due to simplifications or assumptions made by the model

- consistent error across diff dataset.

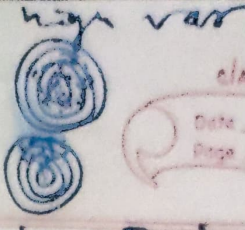
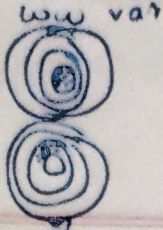
- simplistic assumptions may lead to incorrect predictions.

- may lead to underfitting if model is too simple

- may result in high bias error

low bias

high bias



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K-mean clustering

→ used for partitioning dataset into K distinct, non-overlapping clusters. Here's how it works:-

1) Initialization:-

- Randomly select K data points from datasets as initial centroids. These centroids represent initial cluster centers.

2) Assignment Step:-

- Assign each data point to nearest centroid, based on a dist. metric. The step essentially forms clusters by grouping data pts based on their proximity to centroids.

3) Update Step:-

Recalculate centroids of clusters by taking mean of all datasets assigned to each cluster. The step moves the centroids to center of their respective clusters.

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4) Repeat:-

- Repeat the assignment & update steps until convergence, which occurs when centroids no longer change significantly.

5) Convergence

Once the alg converges, the final centroids represent centers of clusters, & each data point belongs to cluster with nearest centroid.

